

Exploratory Data Analysis of U.S. Domestic Flights (2015–2017)

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Abstract

This report presents a comprehensive exploratory data analysis (EDA) of U.S. domestic flight operations from 2015 to 2017 using the Airline Delay and Cancellation dataset. The analysis examines over 17 million flight records to uncover patterns in flight delays, cancellations, airline performance, airport activity, and route-level behavior. Key findings reveal that departure delays are strongly correlated with arrival delays ($\rho \approx 0.95$), late aircraft delay is the dominant delay cause, and delays accumulate throughout the day due to network propagation effects. The study provides actionable insights into operational delay propagation within the U.S. air transportation network.

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1 Introduction

Air travel is a critical component of modern transportation infrastructure. Flight delays and cancellations impose significant economic costs on airlines, passengers, and the broader economy. Understanding the underlying patterns and causes of these disruptions is essential for improving operational efficiency and passenger experience.

This project performs a systematic exploratory data analysis of U.S. domestic flights between 2015 and 2017 using the *Airline Delay and Cancellation Data (2009–2018)* [1] compiled by Yuanyu Wendy Mu, sourced from the U.S. Bureau of Transportation Statistics.

The analysis addresses the following key questions:

- What are the temporal patterns of flight delays and cancellations?
- How do airlines and airports compare in terms of operational performance?
- What is the relationship between departure delays and arrival delays?
- Which delay causes dominate the system?
- How do delays vary across busy flight routes?

2 Data and Methods

2.1 Data Source

The dataset used in this analysis is the *Airline Delay and Cancellation Dataset (2009–2018)* from Kaggle, compiled by Yuanyu Wendy Mu [1]. It contains comprehensive operational data for U.S. domestic flights reported by the U.S. Department of Transportation’s Bureau of Transportation Statistics. For this project, three yearly files (2015, 2016, 2017) were merged into a single dataframe, yielding 17,111,358 flight records across 28 columns.

2.2 Key Variables

Variable	Type	Description
FL_DATE	Date	Scheduled flight date
OP_CARRIER	Categorical	IATA airline code
OP_CARRIER_FL_NUM	Integer	Flight number
ORIGIN	Categorical	Origin airport IATA code
DEST	Categorical	Destination airport IATA code
CRS_DEP_TIME	Integer	Scheduled departure time (HHMM)
DEP_TIME	Float	Actual departure time (HHMM)
DEP_DELAY	Float	Departure delay in minutes
ARR_DELAY	Float	Arrival delay in minutes
CANCELLED	Binary	Flight cancellation indicator
CANCELLATION_CODE	Categorical	Reason for cancellation (A/B/C/D)
DIVERTED	Binary	Diversion indicator
DISTANCE	Float	Flight distance in miles
CARRIER_DELAY	Float	Delay due to carrier (minutes)
WEATHER_DELAY	Float	Delay due to weather (minutes)
NAS_DELAY	Float	Delay due to National Air System (minutes)
LATE_AIRCRAFT_DELAY	Float	Delay due to late aircraft arrival (minutes)

Table 1: Key variables used in the analysis

The variables can be grouped into the following categories:

- **Flight identifiers:** OP_CARRIER, OP_CARRIER_FL_NUM, ORIGIN, DEST
- **Time-related:** FL_DATE, CRS_DEP_TIME, DEP_TIME, CRS_ARR_TIME, ARR_TIME
- **Delay measurements:** DEP_DELAY, ARR_DELAY, CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, LATE_AIRCRAFT_DELAY
- **Operational metrics:** DISTANCE, AIR_TIME, TAXI_IN, TAXI_OUT, ACTUAL_ELAPSED_TIME, CRS_ELAPSED_TIME
- **Status indicators:** CANCELLED, DIVERTED

2.3 Data Preprocessing

Several preprocessing steps were applied:

- A fully empty column (**Unnamed: 27**) with 100% missing values was identified and removed from the dataset.
- Date columns were converted to datetime format to enable temporal extraction.
- Derived features were created to support structured analysis:

- YEAR, MONTH, DAY_OF_WEEK: extracted from flight date for temporal analysis.
- DEP_HOUR: scheduled departure hour derived from CRS_DEP_TIME.
- TIME_OF_DAY: categorical variable (Late Night, Early Morning, Morning, Afternoon, Evening) based on departure hour.
- IS_DELAYED: binary indicator for flights with arrival delay ≥ 15 minutes.
- DELAY_SEVERITY: categorical severity (Minor/Moderate/Severe) based on arrival delay thresholds.
- MAIN_DELAY_CAUSE: primary delay cause identified as the delay category with the maximum reported minutes.
- ROUTE: concatenation of origin and destination airport codes (ORIGIN-DEST) for route-level analysis.

2.4 Missing Data and Data Quality

2.4.1 Missing Value Analysis

The missing value analysis revealed a clear structural pattern driven by operational reporting rules:

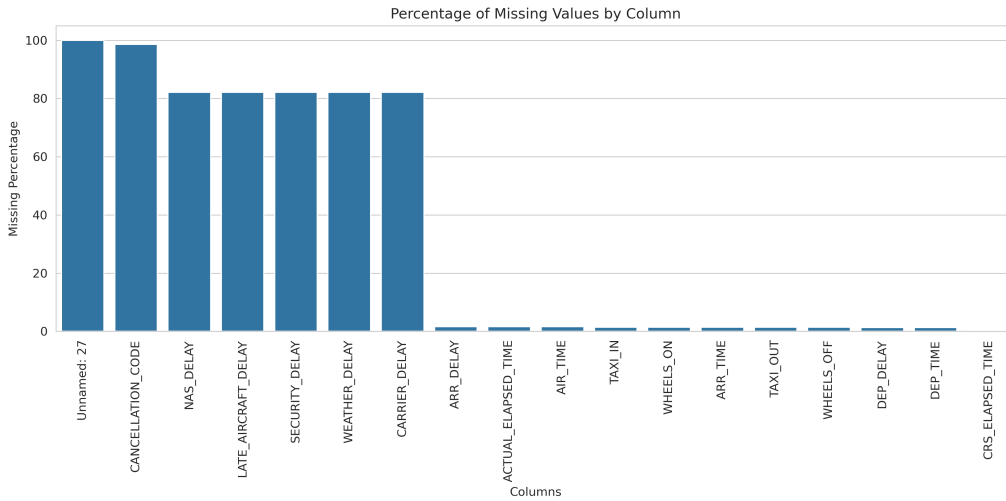


Figure 1: Percentage of missing values across all columns. The **Unnamed: 27** column is entirely empty. Delay cause variables are missing for approximately 82% of records.

Key observations:

- **Unnamed: 27**: 100% missing — removed from the dataset.
- **CANCELLATION_CODE**: 98.6% missing — only populated for cancelled flights (~1.4% of records).
- Delay cause variables (**CARRIER_DELAY**, **WEATHER_DELAY**, **NAS_DELAY**, **SECURITY_DELAY**, **LATE_AIRCRAFT_DELAY**): ~82% missing — populated only for flights with reported delays.
- **ARR_DELAY**: 1.6% missing — absent for cancelled and diverted flights.

- **DEP_DELAY:** minimal missing values.

Further investigation confirmed that missing values are non-random and structurally meaningful: cancellation codes exist only for cancelled flights, arrival delays are missing for cancelled and diverted flights, and delay cause variables are populated only for significantly delayed flights.

2.4.2 Handling Missing Values

Missing values were handled according to the analytical context:

- **Scenario 1 (Delay distribution analysis):** For plotting histograms of `ARR_DELAY`, rows with missing values were dropped using `dropna()`. The missing rate is only $\sim 1.6\%$, so this does not bias the distribution significantly.
- **Scenario 2 (Flight counting by carrier):** All rows were retained regardless of missing arrival delay, because cancelled and diverted flights still represent valid flight operations.

2.5 Methodology

The analysis was conducted using Python 3 with the following libraries:

- **pandas:** data manipulation and aggregation
- **matplotlib:** static visualizations
- **seaborn:** statistical graphics

The methodology follows a structured EDA framework:

1. **Data understanding:** inspecting shape, data types, summary statistics
2. **Data quality assessment:** missing value analysis and structural missingness
3. **Univariate analysis:** distributions of delays, distance, flight volume
4. **Categorical analysis:** carrier and airport frequency distributions
5. **Temporal analysis:** trends across years, months, days, and hours
6. **Carrier and airport comparison:** delay and cancellation performance
7. **Bivariate analysis:** relationships between numerical variables
8. **Correlation analysis:** inter-variable relationships and delay cause breakdown

3 Exploratory Analysis

3.1 Univariate Analysis

3.1.1 Flight Volume

Flight volume remained relatively stable across the three years, with summer months (June–August) showing peak traffic due to vacation travel demand.

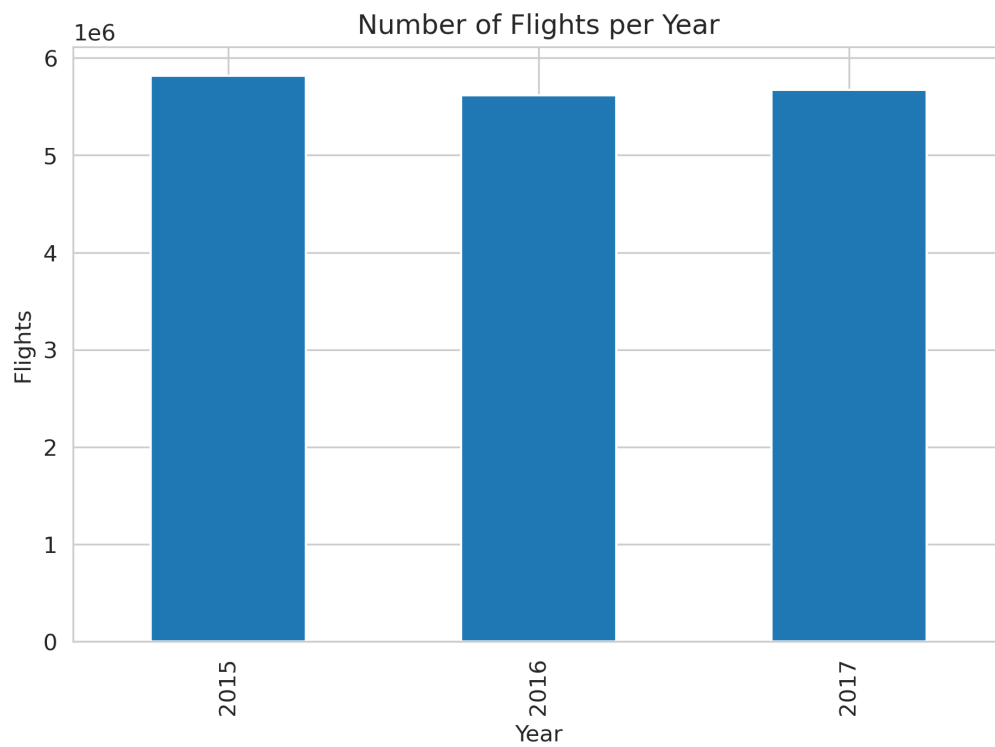


Figure 2: Total number of flights per year (2015–2017).

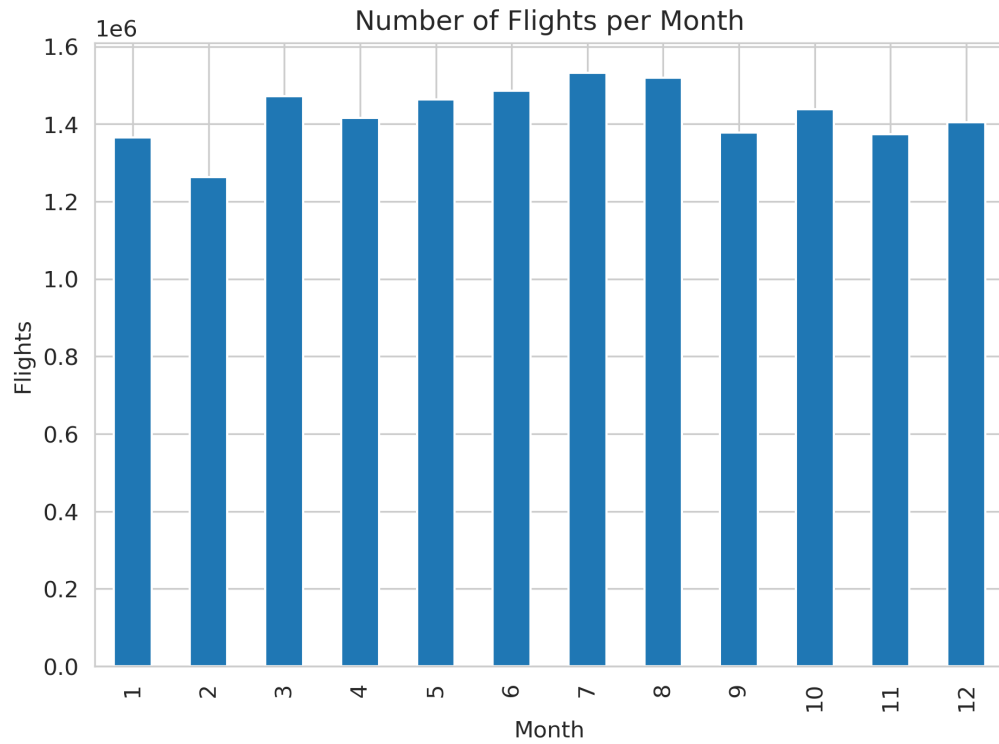


Figure 3: Monthly flight distribution showing peak traffic during summer months (June–August).

3.1.2 Delay Distributions

Both arrival and departure delays exhibit strong right-skewness with long tails. The distributions are centered near zero with median values slightly negative, indicating that most flights arrive or depart slightly ahead of schedule.

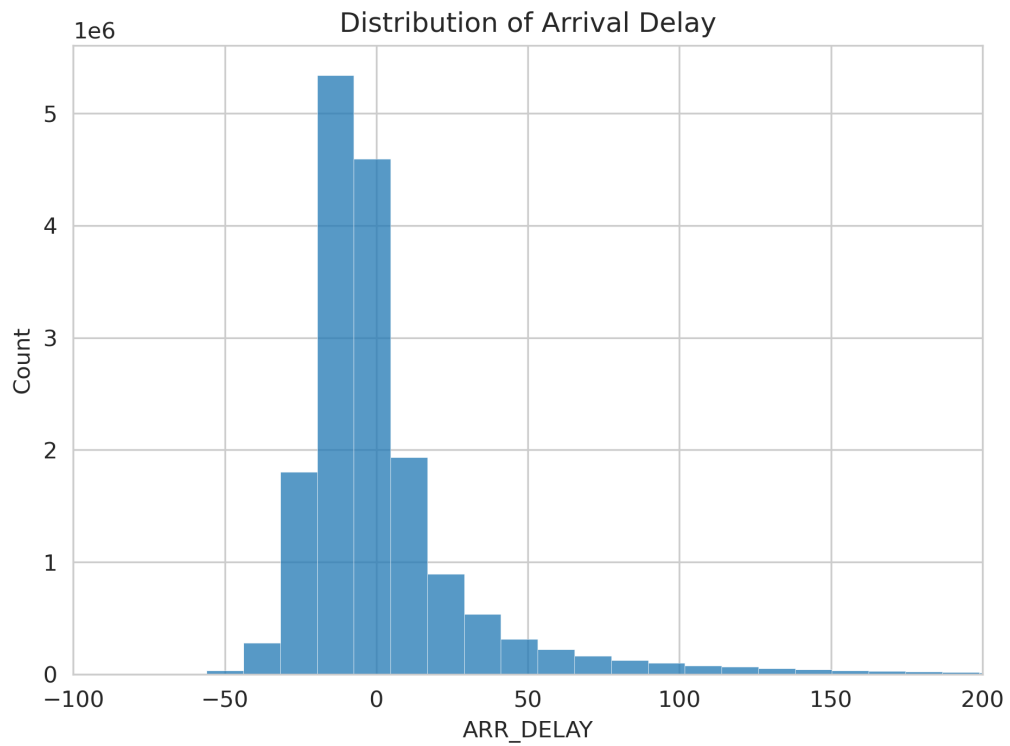


Figure 4: Distribution of arrival delay (minutes).

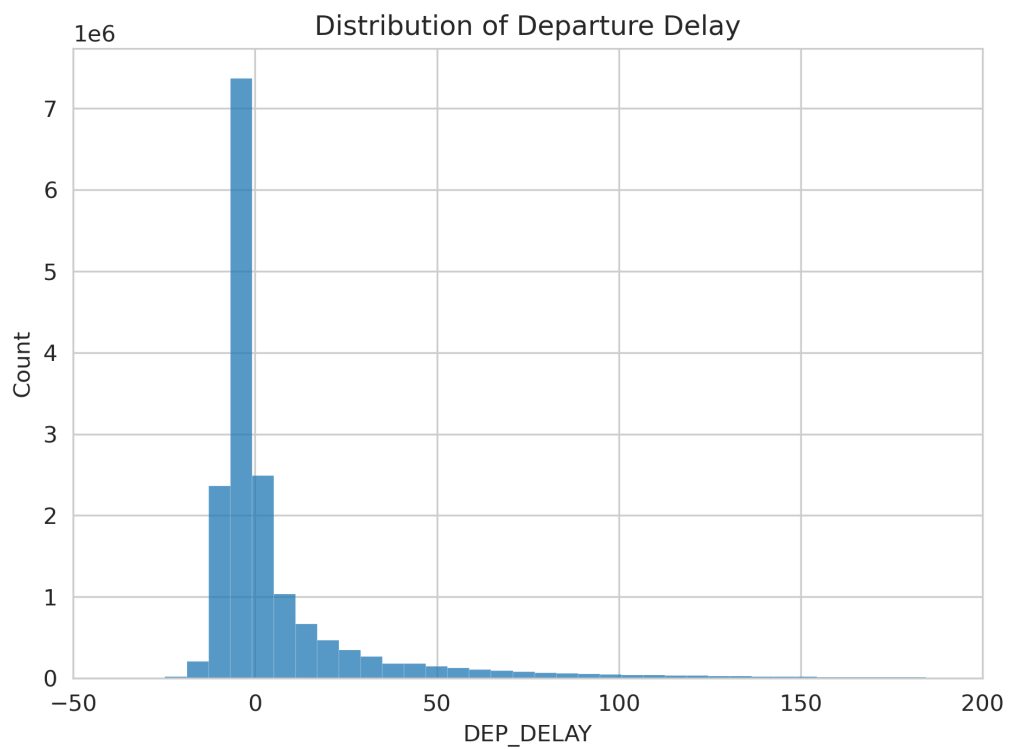


Figure 5: Distribution of departure delay (minutes).

3.1.3 Distance Distribution

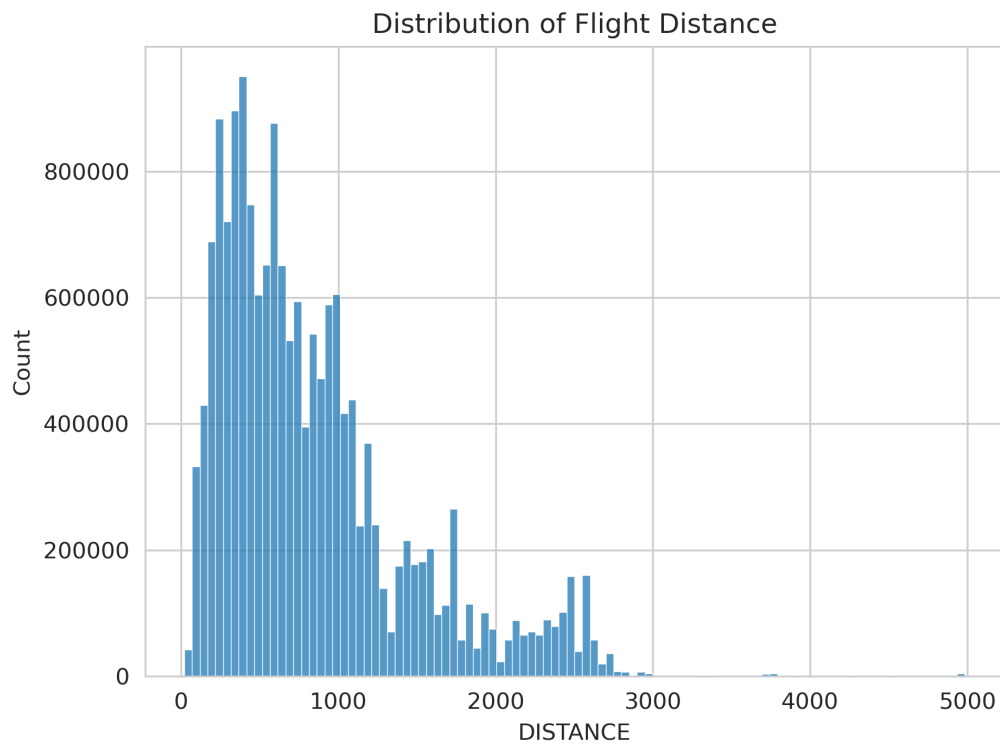


Figure 6: Flight distance distribution. Most flights are short- to medium-haul, concentrated under 2,000 km.

The majority of flights operate on short- to medium-haul routes (under 1,500 km), consistent with the domestic nature of the dataset.

3.2 Categorical Variables

3.2.1 Carrier Distribution

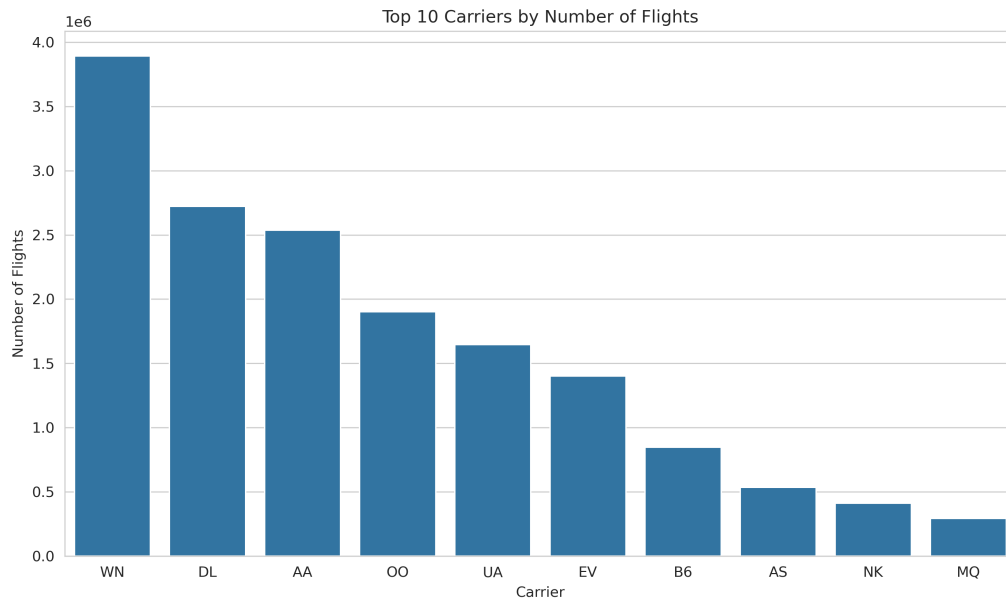


Figure 7: Top 10 carriers by total number of flights. Southwest Airlines (WN) dominates.

The market is concentrated among a few major carriers: WN (Southwest), DL (Delta), and AA (American) account for the largest share of domestic traffic.

3.2.2 Airport Activity

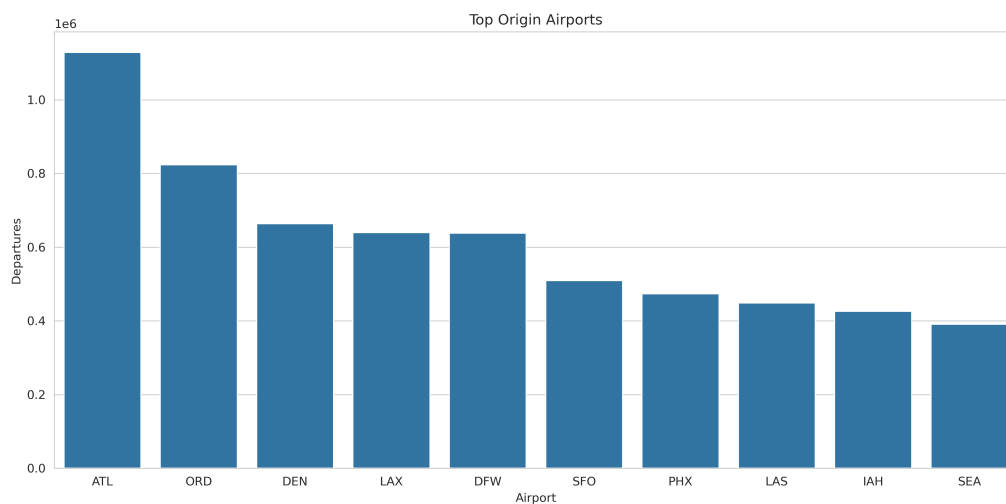


Figure 8: Top 10 busiest origin airports. ATL (Atlanta), ORD (Chicago O'Hare), and DEN (Denver) lead.

Major hub airports dominate departure volumes. ATL handles the highest traffic while maintaining relatively moderate delay levels.

3.2.3 Cancellations and Diversions

The overall cancellation rate is 1.39% and the diversion rate is 0.24%. While small in percentage terms, these represent hundreds of thousands of disrupted flights annually across the three-year period.

3.3 Temporal Patterns

3.3.1 Yearly Trends

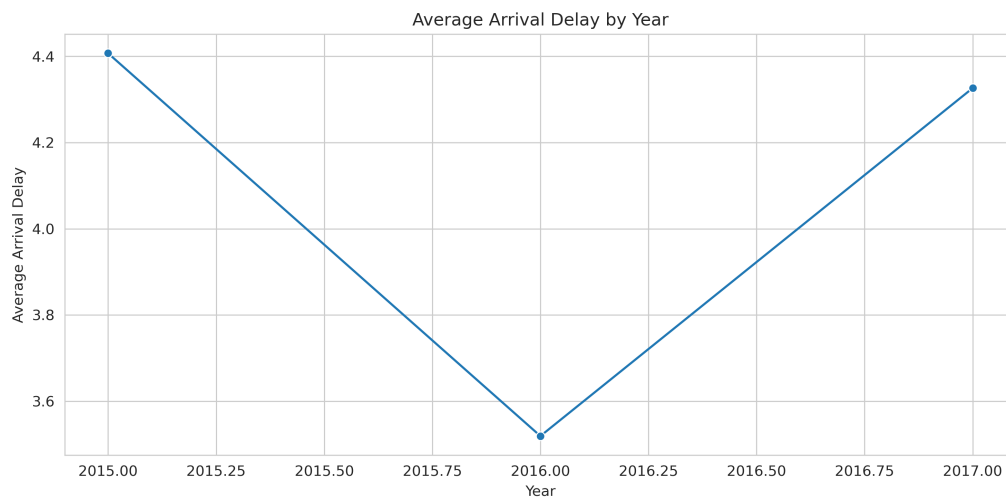


Figure 9: Average arrival delay by year.

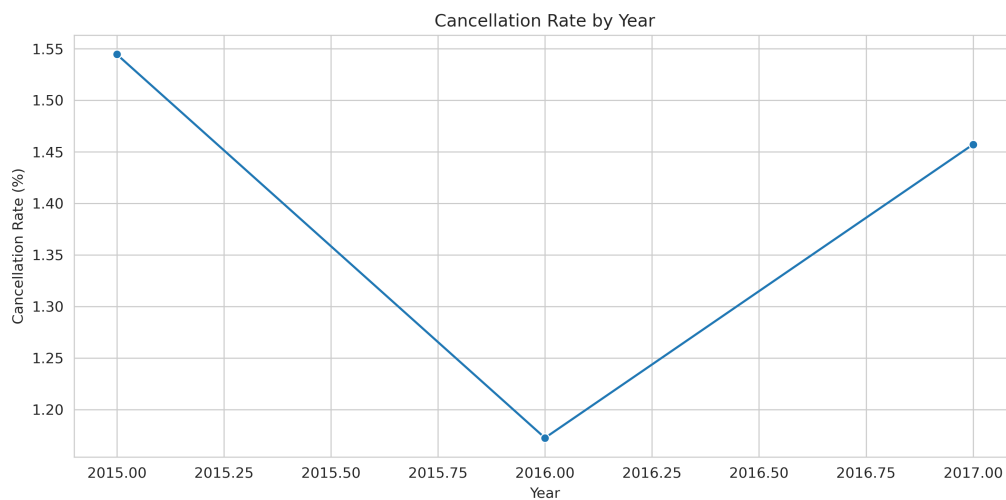


Figure 10: Cancellation rate by year.

3.3.2 Monthly Cancellation Rate

Cancellation rates show clear seasonal variation, with higher rates during winter months (January–February) and summer months (June–August), likely reflecting adverse weather conditions and increased travel demand.

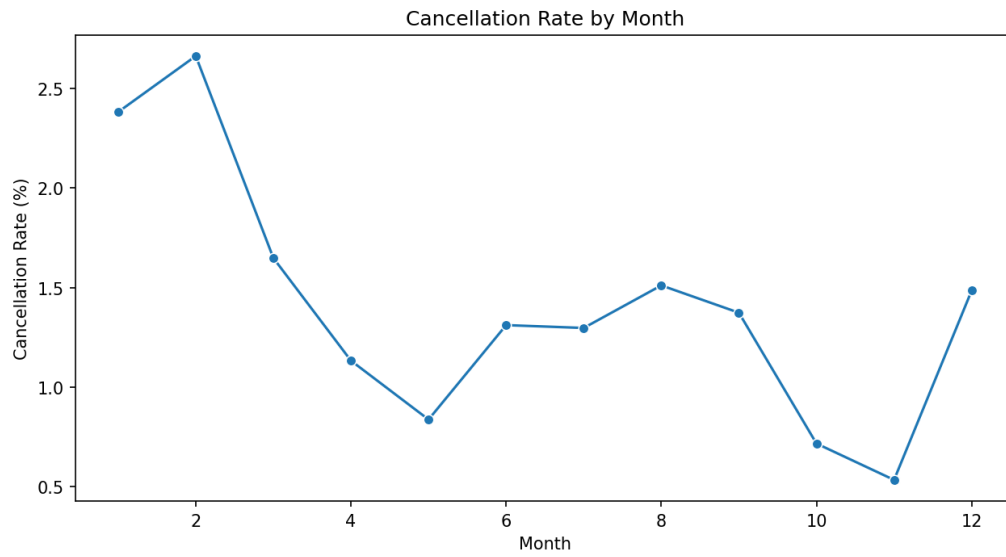


Figure 11: Cancellation rate by month across 2015–2017.

3.3.3 Day of Week and Time of Day

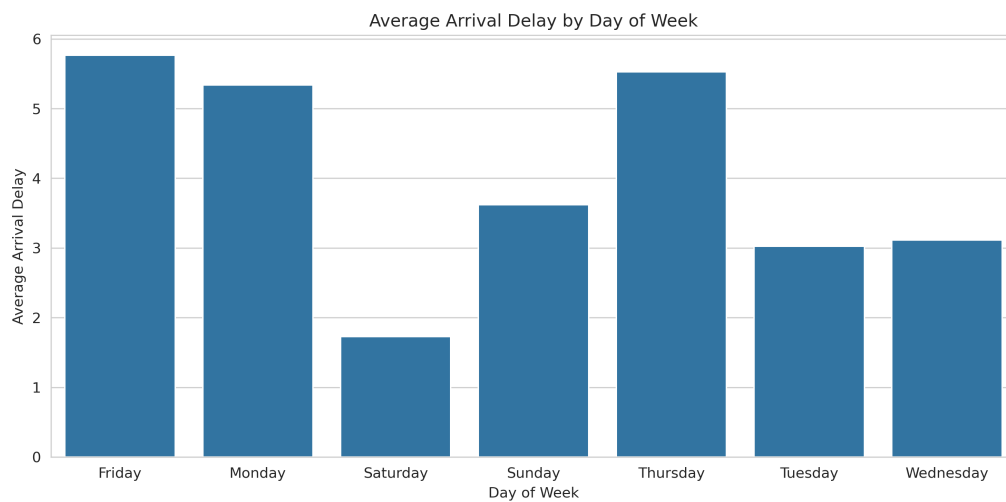


Figure 12: Average arrival delay by day of week. Friday and Monday exhibit the highest average delays.

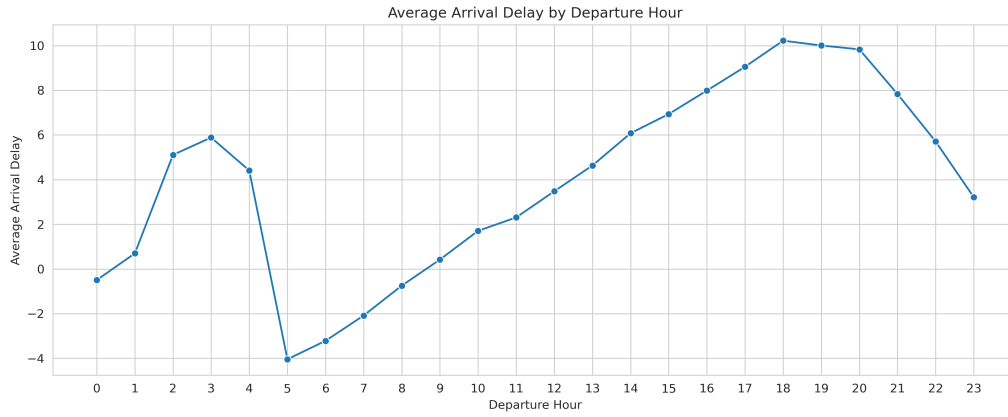


Figure 13: Average arrival delay by scheduled departure hour. Delays increase throughout the day.

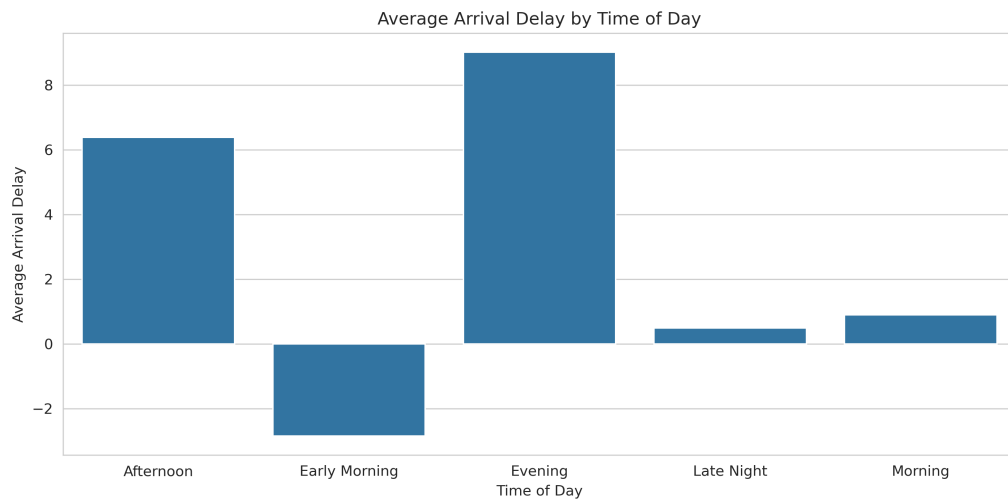


Figure 14: Average arrival delay by time-of-day category.

Key temporal findings:

- **Day of week:** Friday and Monday have the highest average delays, likely due to increased business travel demand.
- **Hour of day:** Delays increase progressively from morning to evening, suggesting delay propagation throughout the day.
- **Seasonality:** Summer months (June–August) and winter months (January–February) have higher cancellation rates.

3.4 Carrier and Airport Performance

3.4.1 Carrier Delay Analysis

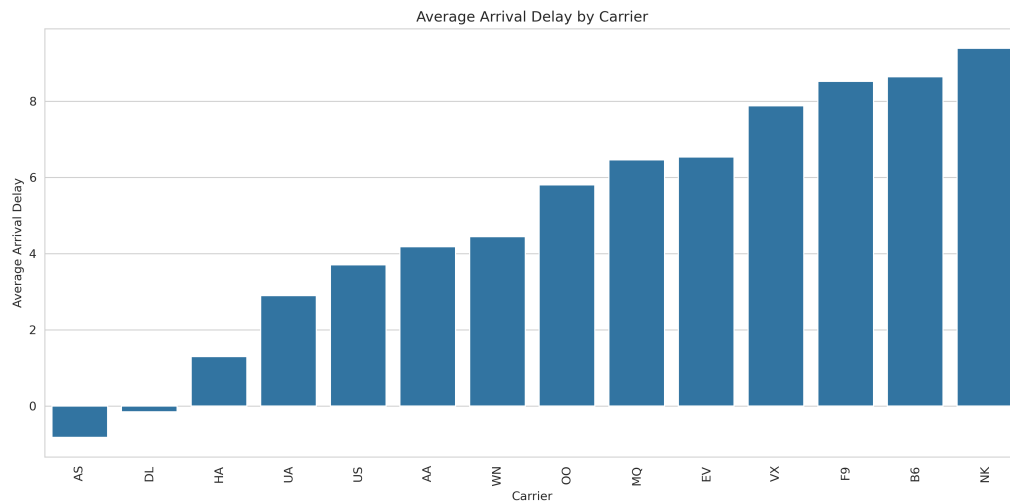


Figure 15: Average arrival delay by carrier. Significant variation exists across airlines.

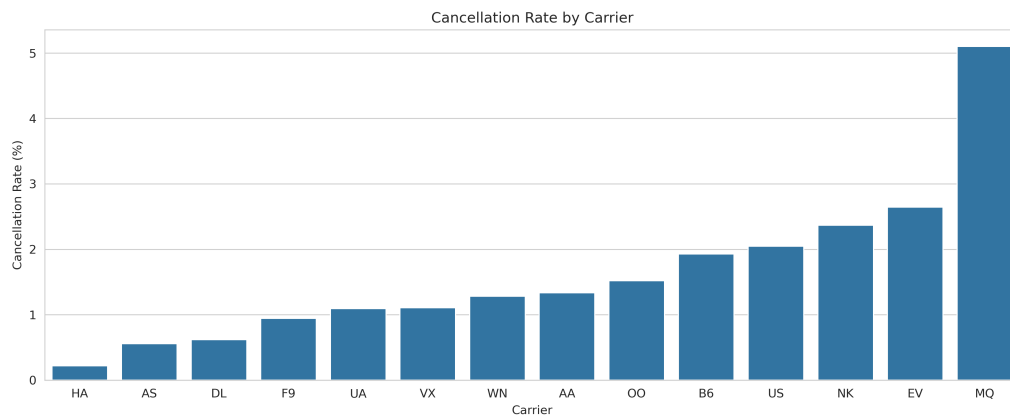


Figure 16: Cancellation rate by carrier. Regional carriers exhibit higher cancellation rates.

Substantial differences are observed across airlines. Hawaiian Airlines (HA), Alaska Airlines (AS), and Delta Air Lines (DL) show the lowest average delays and cancellation rates, while carriers such as Frontier (F9), JetBlue (B6), and Spirit (NK) have higher average delays. Mesa Airlines (MQ) stands out with the highest cancellation rate.

3.4.2 Airport Delay Analysis

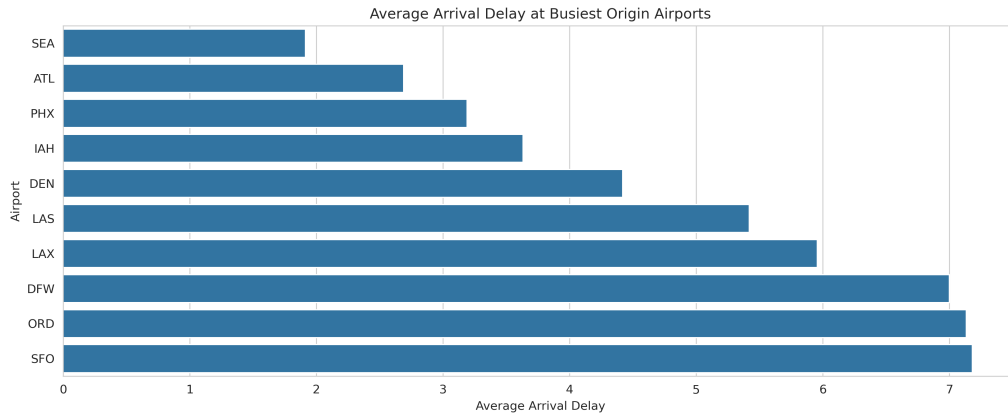


Figure 17: Average arrival delay at the busiest origin airports. Traffic volume alone does not determine delay severity.

Airport-level analysis reveals operational differences among major hubs. ATL handles the highest traffic volume yet maintains moderate delays, while ORD and SFO show higher average delays despite lower traffic, reflecting differences in congestion, weather, and air traffic control efficiency.

3.4.3 Route-Level Analysis

Busy origin-destination routes were analyzed to identify performance patterns beyond individual airports and carriers.

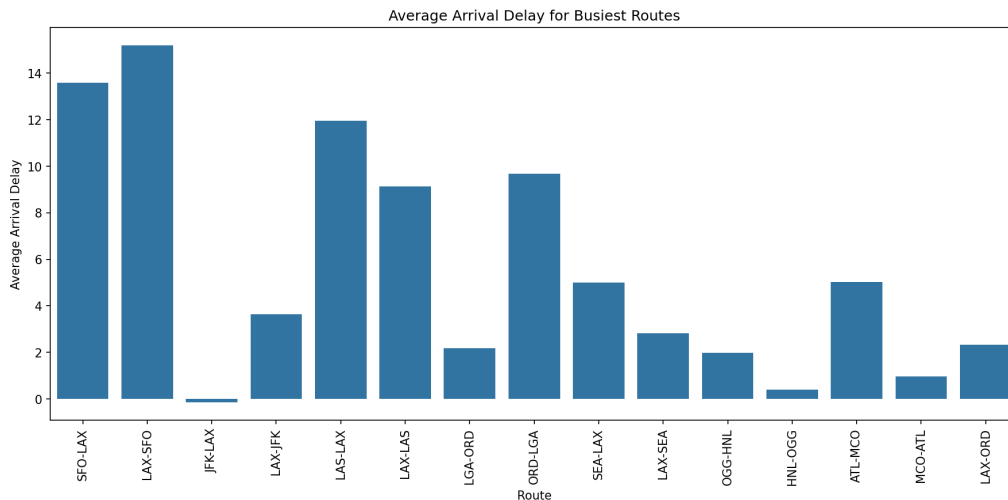


Figure 18: Average arrival delay for the 15 busiest routes. Delay levels vary notably across high-traffic corridors.

The busiest routes predominantly connect major hub airports such as ATL, ORD, DFW, and LAX. Delay levels vary notably across routes, with some high-traffic corridors showing relatively low delays while others experience more frequent disruptions.

3.5 Relationships Between Variables

3.5.1 Departure Delay vs Arrival Delay

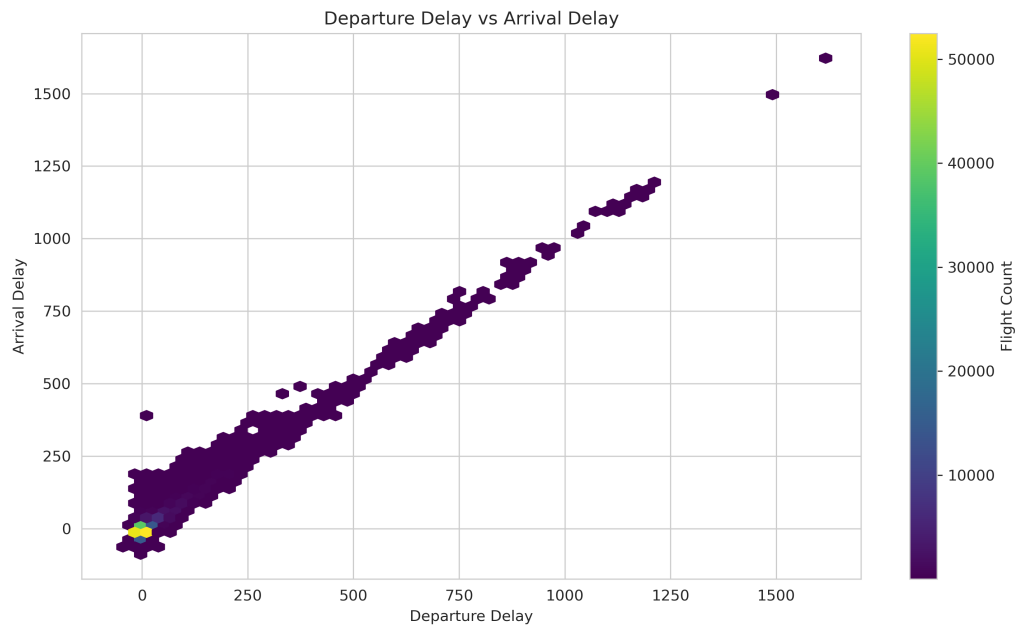


Figure 19: Hexbin density plot of departure delay vs arrival delay. A strong linear relationship is evident ($\rho = 0.95$).

The correlation coefficient between departure delay and arrival delay is approximately 0.95, indicating that departure delays almost entirely propagate to arrival delays. This strong relationship suggests that once a flight departs late, recovering the lost time during the flight is relatively difficult.

3.5.2 Distance vs Arrival Delay

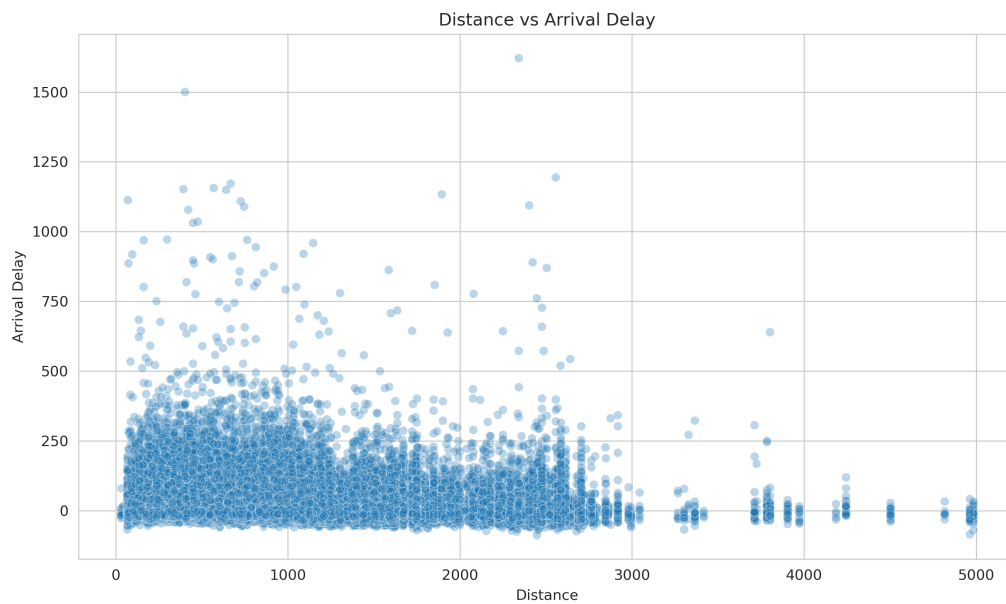


Figure 20: Scatter plot of flight distance vs arrival delay. No linear relationship is observed.

Flight distance shows negligible correlation with arrival delay ($\rho \approx -0.02$). This suggests that longer flights are not inherently more prone to delays, and that operational factors rather than trip length determine delay outcomes.

3.6 Correlation and Causes

3.6.1 Correlation Matrix

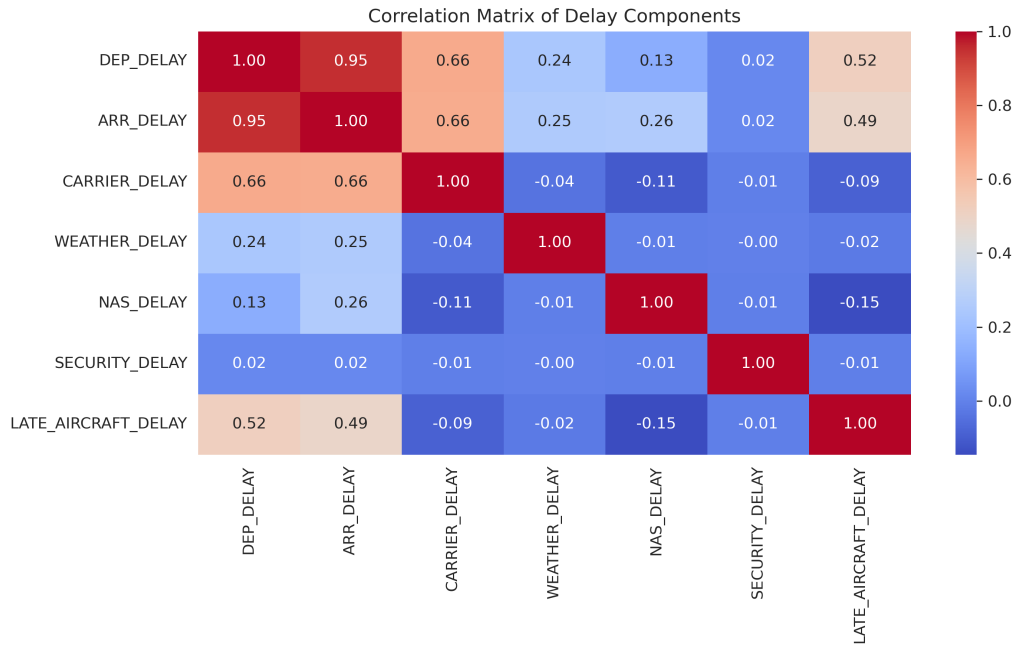


Figure 21: Correlation matrix of delay-related variables. Strong correlations exist between DEP_DELAY, ARR_DELAY, and LATE_AIRCRAFT_DELAY.

The strongest relationships are between departure and arrival delays ($\rho = 0.95$), departure delay and carrier delay ($\rho = 0.66$), and departure delay and late aircraft delay ($\rho = 0.52$). These patterns reinforce the network propagation hypothesis.

3.6.2 Delay Cause Breakdown

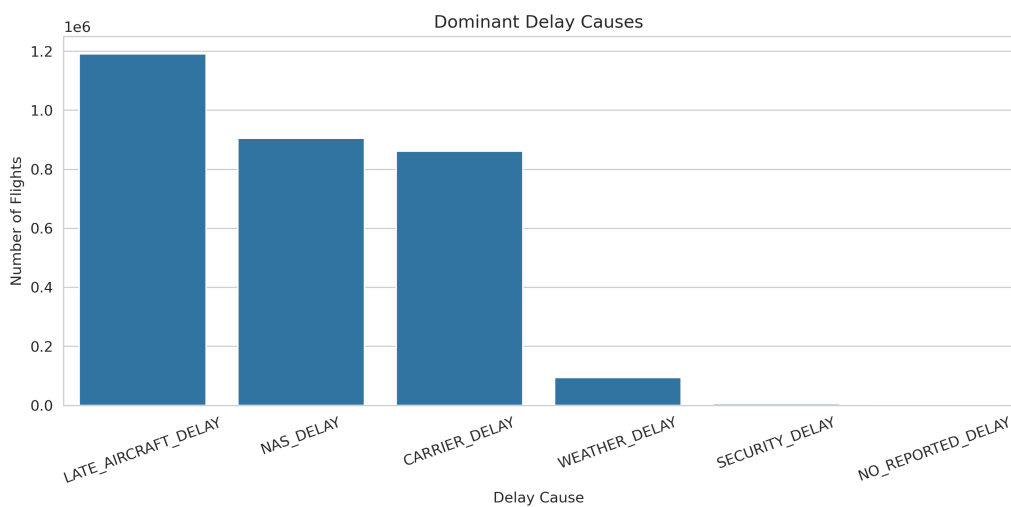


Figure 22: Dominant delay cause for flights with arrival delay ≥ 15 minutes.

Among flights with significant delays (≥ 15 minutes), the most frequently reported primary cause is **late aircraft delay**, followed by NAS delay and carrier delay. This

confirms that delay propagation through the airline network is the primary mechanism behind flight disruptions.

3.6.3 Delay-Cancellation Relationship

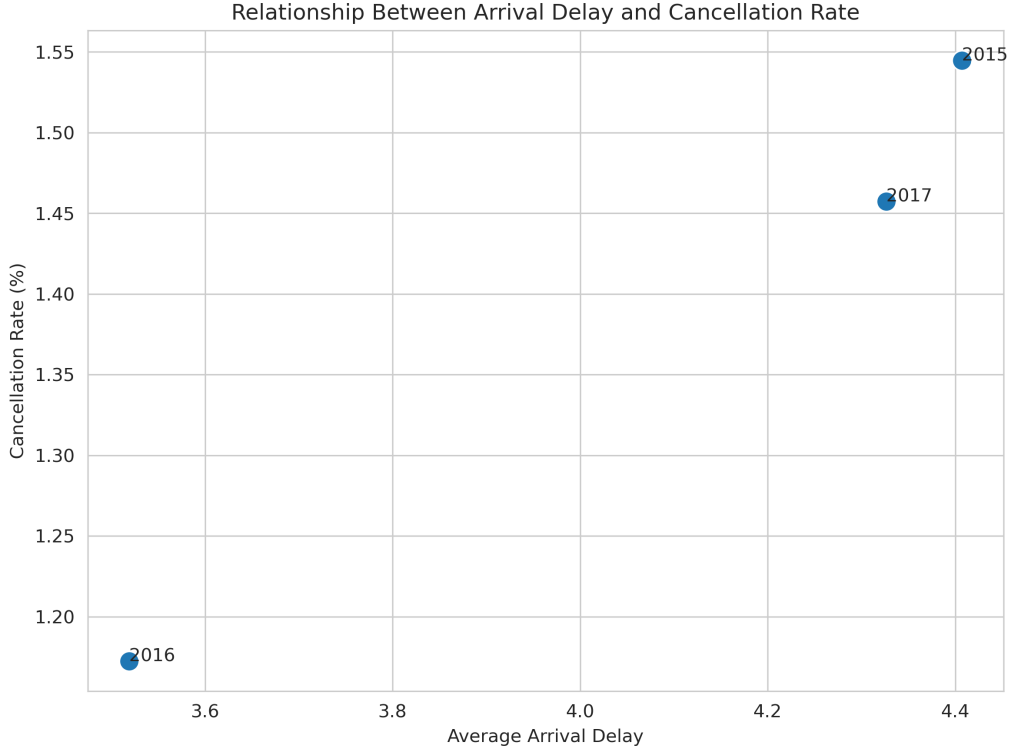


Figure 23: Relationship between average arrival delay and cancellation rate by year. A moderate positive association is observed.

A moderate positive association exists between average arrival delay and cancellation rate at the yearly level, suggesting that years with higher operational disruptions tend to experience more cancellations.

4 Discussion and Limitations

4.1 Key Findings

1. **Delay propagation dominates:** The strong correlation ($\rho = 0.95$) between departure and arrival delays, combined with late aircraft delay being the most common primary cause, indicates that the U.S. domestic flight system is highly interconnected. Delays cascade through the network as aircraft, crew, and schedules are tightly coupled.
2. **Temporal patterns:** Delays accumulate throughout the day, peaking in evening hours. This suggests that early-morning delays, though small, compound over successive flight legs. Cancellation rates follow a seasonal pattern with winter and summer peaks.

3. **Carrier heterogeneity:** Significant variation exists across airlines in both average delay and cancellation rate. This reflects differences in operational strategies, network structures, and schedule buffers.
4. **Airport and route effects:** Traffic volume alone does not predict delay severity. Route-level analysis shows that busy corridors vary significantly in delay performance, suggesting that airport-specific and route-specific factors play important roles.
5. **Distance independence:** Flight distance has minimal impact on delay probability or severity, supporting the network propagation hypothesis over distance-based explanations.

4.2 Limitations

- This analysis is purely descriptive and does not model causal relationships. The correlations identified may reflect confounding factors such as airport congestion, seasonal weather patterns, and airline scheduling practices.
- The dataset covers only 3 years (2015–2017) of the full 2009–2018 range, limiting the ability to detect long-term structural trends.
- Delay cause variables are missing for approximately 82% of flights, which constrains causal analysis of delay origins.
- External data sources such as weather records, air traffic control logs, and airport metadata were not incorporated. Merging with airport location data would enable richer spatial visualizations such as heat maps.
- The dataset does not include passenger-level metrics (e.g., missed connections, compensation claims), limiting the analysis to operational rather than customer-impact dimensions.

5 Bonus Work

5.1 Correlation Analysis of Delay Components

Beyond the standard EDA requirements, a detailed correlation analysis was performed on all seven delay-related variables. The correlation matrix was visualized as a heatmap with annotated coefficients, enabling systematic comparison of how different delay types relate to each other.

5.2 Delay Cause Attribution for Delayed Flights

A novel derived variable (`MAIN_DELAY_CAUSE`) was created to identify the dominant operational cause for each significantly delayed flight (`ARR_DELAY` ≥ 15 minutes). This analysis provides a clearer picture of the most impactful sources of disruption than raw delay minute totals alone.

5.3 Route-Level Performance

An additional layer of analysis was conducted at the route level (origin-destination pairs), examining the 15 busiest domestic corridors. This bridges the gap between carrier-level and airport-level analysis, offering insights into specific high-traffic connections.

6 Conclusion

This exploratory data analysis of 17 million U.S. domestic flights (2015–2017) reveals that airline delays are primarily driven by operational propagation effects rather than external factors such as distance or weather. The near-perfect correlation between departure and arrival delays, the dominance of late aircraft delay as a causal factor, and the accumulation of delays throughout the day all point toward a system where disruptions cascade through interconnected schedules.

These findings have practical implications for airlines and passengers. Schedule optimization, buffer management, and proactive delay mitigation on early-morning flights could reduce the ripple effects that lead to afternoon and evening disruptions. Future work could extend this analysis to the full 10-year dataset, build predictive models for delay classification, incorporate external weather data, and examine the impact of delays on passenger itineraries.

References

- [1] Yuanyu Wendy Mu. (2019). *Airline Delay and Cancellation Data, 2009–2018*. Kaggle. <https://www.kaggle.com/datasets/yuanyuwendymu/airline-delay-and-cancellation-data-2009-2018>
- [2] Ayush Arora. *Kaggle-Airline-Data-Analysis*. GitHub repository. <https://github.com/aroraayush/Kaggle-Airline-Data-Analysis>
- [3] U.S. Department of Transportation, Bureau of Transportation Statistics. *Airline On-Time Performance Data*. <https://www.transtats.bts.gov/>

A Appendix

A.1 Code Repository

The complete analysis code, including data loading, preprocessing, visualization generation, and statistical summaries, is available in the Jupyter notebook `eda.ipynb`. All figures were generated from the notebook and are stored in the `figures/` directory. The full project is hosted on GitHub: <https://github.com/fermow/SBU-Data-Science-2026>

A.2 Full Variable Descriptions

Variable	Description
FL_DATE	Flight date
OP_CARRIER	Operating carrier IATA code
OP_CARRIER_FL_NUM	Flight number
ORIGIN	Origin airport IATA code
DEST	Destination airport IATA code
CRS_DEP_TIME	Scheduled departure time
DEP_TIME	Actual departure time
DEP_DELAY	Departure delay (minutes)
TAXI_OUT	Taxi-out time (minutes)
WHEELS_OFF	Wheels-off time
WHEELS_ON	Wheels-on time
TAXI_IN	Taxi-in time (minutes)
CRS_ARR_TIME	Scheduled arrival time
ARR_TIME	Actual arrival time
ARR_DELAY	Arrival delay (minutes)
CANCELLED	Cancellation indicator
CANCELLATION_CODE	Reason for cancellation
DIVERTED	Diversion indicator
CRS_ELAPSED_TIME	Scheduled elapsed time (minutes)
ACTUAL_ELAPSED_TIME	Actual elapsed time (minutes)
AIR_TIME	Air time (minutes)
DISTANCE	Flight distance (miles)
CARRIER_DELAY	Delay caused by carrier (minutes)
WEATHER_DELAY	Delay caused by weather (minutes)
NAS_DELAY	Delay caused by National Air System (minutes)
SECURITY_DELAY	Delay caused by security (minutes)
LATE_AIRCRAFT_DELAY	Delay caused by late aircraft arrival (minutes)

Table 2: Full variable descriptions

A.3 Technology Stack

- Python 3.13
- pandas 3.0
- matplotlib 3.10
- seaborn 0.13
- L^AT_EX (pdfL^AT_EX)